ADSP 31014 Statistical Models for Data Science
Homework 6

(5 points) Question 1a
Since linear regression is one kind of generalized linear model GLM, it can be estimated using R's `glm()` function too. Let's consider our `chicago_taxi_trip_sample.csv` data set. Fit the linear regression model formula

```
trip_seconds ~ fare + trip_miles + tips_percent
```

using `glm(..., family = gaussian, ...)`. Show the model `summary()`.

(5 points) Question 1b
Fit the same linear regression model formula in Question 1a using the usual `lm()` function. Show the model `summary()` to verify both functions yield the same fitted model.

(5 points) Question 1c
Extract the residual deviance of Question 1a model. Verify it is same as the sum of squares of residuals of Question 1b model.

(10 points) Question 2a
Downtown to Airport: Let's consider the subset of our `chicago_taxi_trip_sample.csv` data set with downtown departure. We wish to investigate the relationship between airport as destination vs the day and time of travel.

```
downtown <- data[data['pickup_area'] == 'Downtown',]
downtown$to_airport <- downtown$dropoff_area %in% c('Midway', 'O\'Hare')
```

Compute the fraction of trips with airport as destination per hour. (Hint: Use `library(dplyr)` and `downtown %>% group_by(hour) %>% ...`) Show a plot for visualization. Which hour has the highest fraction of trips to airport from downtown?

(10 points) Question 2b
Fit the following logistic regression model with `hour` as explanatory variable.

```
glm(to_airport ~ factor(hour) - 1, family = binomial, data = downtown)
```

Show the model `summary()`. How does having the `-1` in the model formula make the model summary easier to interpret?

(5 points) Question 2c
Use Question 2b model coefficients `coef(...)` to compute the predicted probability of airport as destination for each hour.

(5 points) Question 2d
What is the relationship between the predicted probabilities in Question 2c vs the fractions computed in Question 2a?

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(10 points) Question 2e
Fit the following logistic regression model with both hour and day_of_week as explanatory variables. Here `day_of_week = 0` is Monday.

```
glm(to_airport ~ factor(hour) + factor(day_of_week),
    family = binomial, data = downtown)
```

Based on the model `summary()`, which combination of hour and day of week has the highest predicted probability of airport as destination? What is that predicted probability? (Hint: Use `predict(..., newdata=..., type='response')`.)

(5 points) Question 2f
Continuing from Question 2e, which combination of hour and day of week has the lowest predicted probability of airport as destination? What is that predicted probability?

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